

# STATEMENTS AND FUNCTIONS

Building blocks

## STATEMENTS

### Statements

Each instruction we give the computer is a **statement** - executed top to bottom, one after another.

```
int age { 42 }; // a statement that declares a variable and initializes it
greetPerson("Grace"); // a statement that calls a function
return 0; // a statement that ends the program
```

## FUNCTIONS

### A function

A named, reusable block of statements - write the logic once, call it as many times as you like.

```
void greetPerson(std::string name) {  
    std::println("Hello, {}!", name);  
}
```

**name** is a parameter - a placeholder filled in by whoever calls it.

## Compile-time vs. runtime

### Compile-time error

The program won't even build.

```
greetPersonn("Ada");  
// typo: no such function
```

### Runtime error

It builds, but breaks while running.

```
int trouble = numerator / 0;  
// compiles fine, crashes when run
```

## Start the debugging habit now

Breakpoint inside `greetPerson` and inspect `name` - get comfortable doing this before things get complicated.

